```
======= SMART RECORD SEARCHING SYSTEM =======
 1. Add records
 2. Display records
 3. Show sorted copy
 4. Binary search - Iterative
 5. Binary search - Recursive
 6. Compare Binary vs Linear search
 7. Load unsorted sample data
 8. Run test cases
 9. Clear all records
 0. Exit
Enter choice: 1
How many records do you want to enter? 4

--- Record 1 of 4 ---
Enter ID    : 101
Enter Name  : Shreeya Tendolkar
Enter Marks : 90.90

--- Record 2 of 4 ---
Enter ID    : 103
Enter Name  : Diksha pawar
Enter Marks : 91.92

--- Record 3 of 4 ---
Enter ID    : 104
Enter Name  : Gayatri Gawari
Enter Marks : 90.91

--- Record 4 of 4 ---
Enter ID    : 102
Enter Name  : Vaishnavi waghmare
Enter Marks : 90.90
```

```
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 9. Clear all records
 0. Exit
Enter choice: 2

ORIGINAL INPUT ORDER:
Index  ID     Name                 Marks
-----------------------------------------------
0      101    Shreeya Tendolkar    90.90
1      103    Diksha pawar         91.92
2      104    Gayatri Gawari       90.91
3      102    Vaishnavi waghmare   90.90
```

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 9. Clear all records
 0. Exit
Enter choice: 3
[Binary search needs sorted data -> insertion sort applied on a copy; origi
nal input order unchanged]

SORTED COPY (original input is unchanged):
Index  ID     Name                 Marks
-----------------------------------------------
0      101    Shreeya Tendolkar    90.90
1      102    Vaishnavi waghmare   90.90
2      103    Diksha pawar         91.92
3      104    Gayatri Gawari       90.91
```

```
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  1. Add records
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  7. Load unsorted sample data
  8. Run test cases
  9. Clear all records
  0. Exit
 Enter choice: 4
 [Binary search needs sorted data -> insertion sort applied on a copy; origi
 nal input order unchanged]
 Enter ID to search: 103

 Iterative Binary Search trace (on sorted copy):
   Step 1 : low=0  high=3  mid=1  (ID at mid = 102)
   Step 2 : low=2  high=3  mid=2  (ID at mid = 103)
 Result : FOUND at index 2 -> ID=103, Name=Diksha pawar, Marks=91.92 | compa
 risons = 2
 (For duplicate IDs, the first occurrence is returned.)
```

```
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 8. Run test cases
 9. Clear all records
 0. Exit
Enter choice: 5
[Binary search needs sorted data -> insertion sort applied on a copy; origi
nal input order unchanged]
Enter ID to search: 101

Recursive Binary Search trace (on sorted copy):
  Call 1 : low=0  high=3  mid=1  (ID at mid = 102)
  Call 2 : low=0  high=0  mid=0  (ID at mid = 101)
  Call 3 : low=0  high=-1  -> range empty
Result : FOUND at index 0 -> ID=101, Name=Shreeya Tendolkar, Marks=90.90 |
comparisons = 2
(For duplicate IDs, the first occurrence is returned.)
```

```
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 8. Run test cases
 9. Clear all records
 0. Exit
Enter choice: 6
[Binary search needs sorted data -> insertion sort applied on a copy; origi
nal input order unchanged]
Enter ID to search: 102

--- Comparison for ID 102 (n = 4 records) ---
Binary (iterative) [sorted copy] : FOUND at index 1 -> ID=102, Name=Vaishna
vi waghmare, Marks=90.90 | comparisons = 2
Binary (recursive) [sorted copy] : FOUND at index 1 -> ID=102, Name=Vaishna
vi waghmare, Marks=90.90 | comparisons = 2
Linear search      [original   ] : FOUND at index 3 -> ID=102, Name=Vaishna
vi waghmare, Marks=90.90 | comparisons = 4
(Index shown is the position in the array that was searched.)
```

```
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 9. Clear all records
 0. Exit
Enter choice: 7
9 UNSORTED sample records loaded (ID 30 is duplicated).
```

```
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 8. Run test cases
 9. Clear all records
 0. Exit
Enter choice: 8

================ TEST SUITE ================
No  Test case               Key   Expected Iter    Recur   Result
-----------------------------------------------------------------
1   First element           10    0        0       0       PASS
2   Middle element          30    2        2       2       PASS
3   Last element            50    4        4       4       PASS
4   Absent key              35    -1       -1      -1      PASS
5   Single item - present   42    0        0       0       PASS
6   Single item - absent    7     -1       -1      -1      PASS
7   Duplicates (first idx)  30    2        2       2       PASS
8   Empty list              5     -1       -1      -1      PASS
9   Even number of items    6     2        2       2       PASS
10  Reverse-sorted input    1     0        0       0       PASS
-----------------------------------------------------------------
Passed 10 / 10 test cases.
```

```
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 8. Run test cases
 9. Clear all records
 0. Exit
Enter choice: 9
All records cleared.
```

```
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 6. Compare Binary vs Linear search
 7. Load unsorted sample data
 8. Run test cases
 9. Clear all records
 0. Exit
Enter choice: 0
Goodbye!
```